

Data Cleaning, Exploratory Data Analysis (EDA) & Basic Business Reporting

1. Introduction

This project focuses on Data Cleaning, Exploratory Data Analysis (EDA), SQL analysis, and Basic Business Reporting using the Superstore Sales Dataset.

The objective of this project is to transform raw sales data into meaningful business insights using Excel, PostgreSQL, and Power BI.

The project covers data cleaning, duplicate checking, sales analysis, KPI creation, SQL-based analysis, and interactive dashboard development.

2. Objective

The main objectives of this project are:

- To clean and prepare the sales dataset for analysis.
- To identify and check missing or duplicate records.
- To perform exploratory data analysis using Excel Pivot Tables.
- To analyze sales based on categories, regions, and sub-categories.
- To perform basic SQL analysis using PostgreSQL.
- To create an interactive Power BI dashboard.
- To calculate important business KPIs.
- To generate useful business insights from the analyzed data.

3. Dataset

The project uses the Superstore Sales Dataset.

The dataset contains sales-related information such as:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- Country
- City

- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales

The dataset contains approximately 9,800 sales records.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Name	Segment	Country	City	State	Postal	Region	Product ID	Category	Sub-Categ	Product Name
1	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
2	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
3	CA-2017-138688	12-06-2017	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal
4	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
5	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
6	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
7	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
8	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
9	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsil
10	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C20VTELE Outlet Surge
11	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables
12	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 ConferenceA phoneA Charcoal black
13	CA-2018-114412	13-04-2018	20-04-2018	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Car	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
14	CA-2017-161389	05-12-2017	10-12-2017	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes P8200 Plastic Comb Binding Machine
15	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AR-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, H
16	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders
17	CA-2015-105893	11-11-2015	18-11-2015	Standard Class	PK-19675	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
18	CA-2015-167184	13-05-2015	13-05-2015	Second Class	AG-10170	Aljandro Grove	Consumer	United States	West Jordan	Utah	84004	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer
19	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341
20	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone
21	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"
22	CA-2017-137330	09-12-2017	13-12-2017	Standard Class	KB-16085	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art	Newell 318
23	CA-2017-137330	09-12-2017	13-12-2017	Standard Class	KB-16085	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Appliances	Acco Six-Outlet Power Strip, 4' Cord Length
24	US-2018-156909	16-07-2018	18-07-2018	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylv	19140	East	FUR-CH-10002774	Furniture	Chairs	Global Deluxe Stacking Chair, Gray
25	CA-2016-106320	25-09-2016	30-09-2016	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
26	CA-2017-121755	16-01-2017	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders	Wilson Jones Active Use Binders
27	CA-2017-121755	16-01-2017	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	ImationA 8GB Mini TravelDrive USB 2.0A Flash Drive
28	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish
29	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	OFF-BI-10000474	Office Supplies	Binders	Avery Recycled Flexi-View Covers for Binding Systems
30	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	FUR-FU-10004848	Furniture	Furnishings	Howard Miller 13-3/4" Diameter Brushed Chrome Round Wall Clock
31	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	OFF-EN-10001509	Office Supplies	Envelopes	Poly String Tie Envelopes
32	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	OFF-AR-10004042	Office Supplies	Art	BOSTON Model 1800 Electric Pencil Sharpeners, Putty/Woodgrain
33	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	OFF-BI-10001525	Office Supplies	Binders	Acco Pressboard Covers with Storage Hooks, 14 7/8" x 11", Executive F
34	US-2016-150630	17-09-2016	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylv	19140	East	OFF-AR-10001683	Office Supplies	Art	Lumber Crayons

4. Data Cleaning Using Excel

The dataset was first imported into Microsoft Excel for data cleaning and preparation.

The following activities were performed:

4.1 Data Filtering

Filters were applied to the dataset to inspect the available values and identify possible blank or missing entries.

4.2 Duplicate Check

The Remove Duplicates feature was used to check the dataset for duplicate records.

The analysis showed that no duplicate records were found.

4.3 Data Format Verification

The date and numerical fields were checked to ensure that they were available in suitable formats for analysis.

The Order Date and Ship Date fields were verified as date-related fields, while Sales was checked as a numerical field.

4.4 Data Preparation

After checking the dataset, the cleaned data was prepared for further analysis using Excel, SQL, and Power BI.

5. Exploratory Data Analysis Using Excel

Pivot Tables were created in Excel to understand sales performance across different business dimensions.

5.1 Category-wise Sales Analysis

The Category field was used as Rows and Sales was used as Values in the Pivot Table.

The results were:

- Technology: 827,455.87
- Furniture: 728,658.58
- Office Supplies: 705,422.33

Technology generated the highest sales among the three categories in the analyzed dataset.

5.2 Region-wise Sales Analysis

A Region-wise Sales Pivot Table was also created.

The results were:

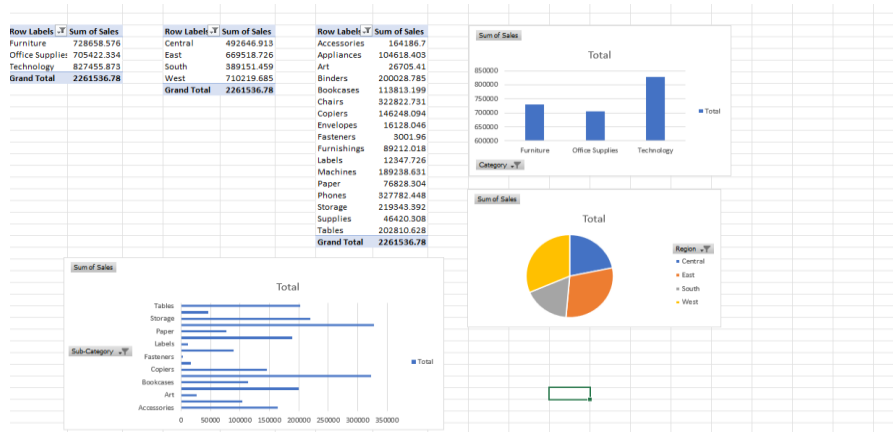
- West: 710,219.68
- East: 669,518.73
- Central: 492,646.91
- South: 389,151.46

The West region recorded the highest sales, while the South region recorded the lowest sales among the four regions.

5.3 Sub-Category Analysis

A Pivot Table was created using Sub-Category and Sales to analyze sales performance across individual product sub-categories.

This analysis helps identify high-performing and low-performing product segments.



6. SQL Analysis Using PostgreSQL

The cleaned sales dataset was imported into PostgreSQL for SQL-based analysis.

A separate table named sales_data was created for the project.

SQL concepts used during the analysis included:

- SELECT
- GROUP BY
- ORDER BY
- SUM()
- AVG()
- COUNT()
- DISTINCT

SQL was used to perform structured analysis of the sales data and support the business reporting process.

Showing rows: 1 to 1000 Page No: 1 of 10								
	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
1	1	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consum
2	2	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consum
3	3	CA-2017-138688	12-06-2017	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporat
4	4	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consum
5	5	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consum
6	6	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
7	7	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
8	8	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
9	9	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
10	10	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
11	11	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
12	12	CA-2015-115812	09-06-2015	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consum
13	13	CA-2018-114412	15-04-2018	20-04-2018	Standard Class	AA-10480	Andrew Allen	Consum
14	14	CA-2017-161389	05-12-2017	10-12-2017	Standard Class	IM-15070	Irene Maddox	Consum
15	15	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home O
16	16	US-2016-118983	22-11-2016	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home O
17	17	CA-2015-105893	11-11-2015	18-11-2015	Standard Class	PK-19075	Pete Kriz	Consum
18	18	CA-2015-167164	13-05-2015	15-05-2015	Second Class	AG-10270	Alejandro Grove	Consum
19	19	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consum
20	20	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consum
21	21	CA-2015-143336	27-08-2015	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consum

7. Power BI Dashboard

The sales data was imported into Power BI to create an interactive business dashboard.

The dashboard contains different visualizations for analyzing sales performance.

Visualizations Created

1. Region-wise Sales – Pie Chart
2. Category-wise Sales – Clustered Bar Chart
3. Sub-Category Sales – Clustered Bar Chart

These visualizations make it easier to understand sales distribution across different regions, categories, and sub-categories.

8. Key Performance Indicators (KPIs)

The following KPI cards were created in Power BI:

Total Sales

2.6 Million

Total Orders

4,922

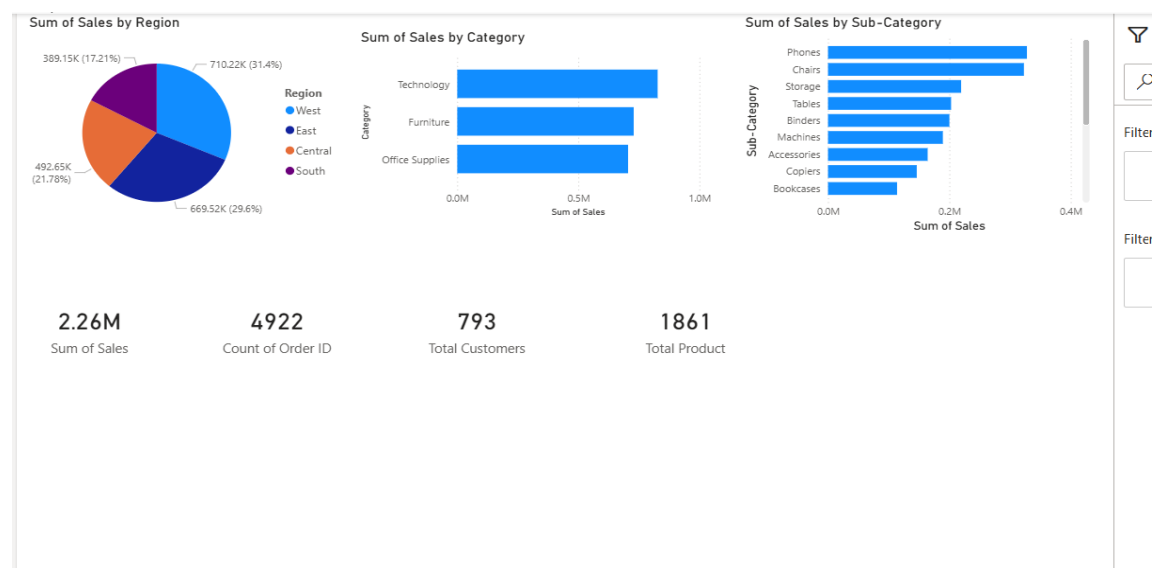
Total Customers

793

Total Products

1,861

These KPIs provide a quick overview of the overall sales dataset and business performance.



9. Business Insights

Based on the analysis performed, the following insights were identified:

- Technology generated the highest sales among the analyzed product categories.
- Furniture generated the second-highest category sales.
- Office Supplies recorded the lowest sales among the three main categories.
- The West region recorded the highest sales.
- The East region was the second-highest performing region.
- The South region recorded the lowest sales among the four regions.
- The dashboard provides a quick overview of sales performance through KPI cards.
- Category, region, and sub-category analysis can help businesses understand sales distribution and identify areas requiring further investigation.

10. Tools and Technologies Used

- Microsoft Excel
- Excel Pivot Tables
- PostgreSQL
- SQL
- Power BI
- Data Cleaning
- Exploratory Data Analysis
- Business Intelligence
- Data Visualization

11. Conclusion

This project provided practical experience in Data Analytics and Business Intelligence.

The Superstore dataset was cleaned and analyzed using Excel, SQL analysis was performed using PostgreSQL, and an interactive Power BI dashboard was created for business reporting.

The project helped in understanding how raw sales data can be transformed into structured information, KPIs, visual reports, and meaningful business insights.

Overall, this task strengthened practical skills in data cleaning, exploratory data analysis, SQL, Power BI, dashboard development, and business reporting.